WEAK ORBITS AGREE WITH ORBITS UNDER LOCALLY FINITE TYPE OR LARGE CARDINALITY HYPOTHESES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file groupoids-quotients.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The cardinality bound lambda(R) is the one introduced just before the cited large-enough lemma.

Editorial note. The label lemma-geometric-orbits names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Chapter Lemma lemma-make-pre-equivalence (navigation stub, not delivered): replacing a pre-relation R by the pre-equivalence relation R-infinity it generates leaves weak orbits, orbits, and invariant morphisms unchanged, so orbit statements may be proven for pre-equivalence relations
- Chapter Lemma lemma-weak-orbit-pre-equivalence (navigation stub, not delivered): for a pre-equivalence relation, R-equivalence of k-valued points over extension fields is expressed by the image of R(k) in U(k) times U(k)
- Morphisms of Spaces, Lemma spaces-morphisms-lemma-permanence-finite-type: local finite type is preserved under composition
- Morphisms of Spaces, Lemma spaces-morphisms-lemma-base-change-finite-type: local finite type is preserved under base change
- Morphisms of Spaces, Lemma spaces-morphisms-lemma-locally-finite-type-surjective-geometric-points: a nonempty algebraic space locally of finite type over an algebraically closed field has a k-valued point
- Morphisms of Spaces, Lemma spaces-morphisms-lemma-large-enough: the same conclusion holds over any algebraically closed field whose cardinality exceeds the bound lambda associated to the morphism

Source reference. Original reference 048T: sources/stacks-groupoids-quotients/pinned-original-source.tex, source lines 866–866. Prerequisite pointer invoked at 908; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 048R: sources/stacks-groupoids-quotients/pinned-original-source.tex, source lines 781–781. Prerequisite pointer invoked at 915; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:groupoids-quotients.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 03XH: sources/stacks-chow/tags.txt, source lines 5148–5148. Prerequisite pointer invoked at 926; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0488: sources/stacks-chow/tags.txt, source lines 5535–5535. Prerequisite pointer invoked at 937; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0487: sources/stacks-chow/tags.txt, source lines 5534–5534. Prerequisite pointer invoked at 929; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0462: sources/stacks-chow/tags.txt, source lines 5457–5457. Prerequisite pointer invoked at 922; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme, and let B be an algebraic space over S. Let $j : R \to U \times_B U$ be a pre-relation over B. Let $\mathrm{Spec}(k) \to B$ be a geometric point of B.*

1. *If $s, t : R \to U$ are locally of finite type then weak R-equivalence on $U(k)$ agrees with R-equivalence, and weak R-orbits agree with R-orbits on $U(k)$.*
2. *If k has sufficiently large cardinality then weak R-equivalence on $U(k)$ agrees with R-equivalence, and weak R-orbits agree with R-orbits on $U(k)$.*

Proof. We first prove (1). Assume s, t locally of finite type. By Lemma 048T we may assume that R is a pre-equivalence relation. Let k be an algebraically closed field over B. Suppose $\overline{u}, \overline{u}' \in U(k)$ are R-equivalent. Then for some extension field Ω/k there exists a point $\overline{r} \in R(\Omega)$ mapping to $(\overline{u}, \overline{u}') \in (U \times_B U)(\Omega)$, see Lemma 048R. Hence

$$Z = R \times_{j, U \times_B U, (\overline{u}, \overline{u}')} \mathrm{Spec}(k)$$

is nonempty. As s is locally of finite type we see that also j is locally of finite type, see Morphisms of Spaces, Lemma 0462. This implies Z is a nonempty algebraic space locally of finite type over the algebraically closed field k (use Morphisms of Spaces, Lemma 03XH). Thus Z has a k-valued point, see Morphisms of Spaces, Lemma 0487. Hence we conclude there exists a $\overline{r} \in R(k)$ with $j(\overline{r}) = (\overline{u}, \overline{u}')$, and we conclude that $\overline{u}, \overline{u}'$ are R-equivalent as desired.

The proof of part (2) is the same, except that it uses Morphisms of Spaces, Lemma 0488 instead of Morphisms of Spaces, Lemma 0487. This shows that the assertion holds as soon as $|k| > \lambda(R)$ with $\lambda(R)$ as introduced just above Morphisms of Spaces, Lemma 0487. □